

Geography Ch-17 (Advanced): India's Savanna-Type Lands - Tropical Dry Deciduous Forests & Grasslands

Tier 2: Advanced | Source: D.R. Khullar + Majid Husain (Geography of India)
India-applied grounded in Qdrant upsc_static book DB + live Current Affairs anchor

Facts from official sources | Inferences clearly marked | No fabricated data

HIGH

India's Savanna Analogue: Tropical Dry Deciduous Forests, Thorn Scrub & Grasslands

GS-I (Physical Geography of India / Biogeography) Geography

■ NEWS TRIGGER

Advanced application (D.R. Khullar; Majid Husain). India's savanna-equivalent is the tropical dry deciduous forest and thorn/scrub belt of the peninsular interior and NW - shaped by a strong wet-dry monsoon rhythm just like the Sudan climate.

■ INTRO & ORIGIN

Much of peninsular India falls under Koppen Aw (savanna). Its natural cover - the tropical dry deciduous forest (monsoon forest), grading into thorn forest and scrub in drier tracts - is India's savanna analogue: trees shed leaves in the dry season, grasses dominate openings, and fire + grazing keep the landscape open.

Where SW-monsoon rain is moderate (~75-150 cm) and the dry season pronounced, the wet-dry rhythm produces deciduous ('monsoon') forest; below ~75 cm it degrades to thorn scrub - India's driest savanna margin.

■ KEY DATA

Type	Rainfall	Region / species
Tropical moist deciduous	100-200 cm	Teak, sal; Ghats' leeward, NE peninsula
Tropical dry deciduous	70-100 cm	Teak, tendu, axlewood; central India, Deccan
Tropical thorn / scrub	< 70 cm	Acacia, babul, khejri; Rajasthan, Deccan rain-shadow
Tropical/montane grasslands	Varies	Terai, shola grasslands, Banni (Kachchh)

■ STATIC THEORY

- **Tropical deciduous ('monsoon') forests** are India's most widespread type; they shed leaves for 6-8 weeks in the dry season to conserve water - the savanna-type response. Split into MOIST deciduous (100-200 cm; sal, teak) and DRY deciduous (70-100 cm; teak, tendu, axlewood).
- **Tropical thorn forest & scrub** (< 70 cm) occupies the Thar margins, western Deccan rain-shadow and parts of Gujarat/Punjab; species are xerophytic - acacia, babul, khejri (Prosopis), euphorbia; long roots, thorns, waxy/reduced leaves.
- **Indian grasslands (savanna analogues)**: the Terai (Himalayan foothills, tall wet grassland - home of the one-horned rhino), the shola grasslands of the high Western Ghats, and the Banni grassland of Kachchh (Asia's largest, home of the Indian wild ass nearby in the Rann).
- **Fire & grazing**: as in the African savanna, fire and grazing keep these Indian grasslands and open forests from reverting to closed forest.
- **Soils - laterite risk**: the pronounced wet-dry regime promotes leaching and laterite formation on interfluvies - poor for agriculture unless managed, echoing GC Leong's savanna soil-degradation point.

■ MUST-KNOW FACTS

1. India's savanna analogue = tropical dry deciduous forest + thorn scrub + grasslands (peninsular interior & NW).
2. Moist deciduous (100-200 cm; sal, teak) vs dry deciduous (70-100 cm) vs thorn (< 70 cm; acacia, khejri).

■ UPSC TRAPS

✗ India's dry deciduous forests are evergreen.

✓ They are DECIDUOUS - they shed leaves in the dry season (a savanna-type adaptation); evergreen forests are on the wet windward Ghats/NE.

3. Deciduous trees shed leaves in the dry season - the classic savanna water-conservation strategy.
4. Key Indian grasslands: Terai (rhino), shola (W. Ghats), Banni (Kachchh - Asia's largest).
5. Wet-dry regime -> leaching -> laterite soils on interfluvies (agricultural challenge).

✗ Grasslands like Banni and Terai are the same type.

✓ Terai = tall WET grassland of the humid foothills;
Banni = arid/saline grassland of Kachchh - very different moisture regimes.

■ MAINS ANGLE

Explain how India's tropical dry deciduous forests and grasslands function as the country's savanna analogue, and why fire, grazing and laterite soils make them ecologically fragile.

■ STUDY LINK

Geography -> Natural Vegetation of India -> Deciduous/Thorn forests & grasslands; links to savanna (Aw) climate

■ MEDIUM UPSC RELEVANCE

MEDIUM

CA Anchor: India's 'Aravalli Green Wall' Project Against Desertification

GS-I (Geography of India) + GS-III (Environment / Land Degradation)
Geography - Current Affairs

■ NEWS TRIGGER

The Aravalli Green Wall Project (launched 2023, MoEFCC) aims to create a green buffer along the ~1,400 km Aravalli range across Haryana, Rajasthan, Gujarat and Delhi (with a ~5 km buffer) - India's own 'green wall' to check the eastward march of the Thar desert, echoing Africa's GGW.

■ INTRO & ORIGIN

India faces its own savanna-margin desertification: the Aravallis are the natural barrier stopping the Thar from spreading into the fertile Indo-Gangetic plains. The Aravalli Green Wall Project is the applied Indian counterpart to the African Great Green Wall.

Part of India's commitment to restore 26 million ha of degraded land by 2030 (UNCCD/Bonn Challenge). ISRO's Desertification Atlas shows ~30% of India's land degraded (link to Ch-7 Advanced).

■ KEY DATA

Aspect	Detail	Purpose
Launch	2023, MoEFCC	Green buffer along Aravallis
Extent	~1,400 km + ~5 km buffer	Haryana, Rajasthan, Gujarat, Delhi
Goal	Afforestation, water-body restoration	Halt Thar's eastward spread
Global link	Bonn Challenge / UNCCD	Restore 26 mn ha by 2030

■ STATIC THEORY

- **Aravalli range:** among the world's oldest fold mountains; act as a natural ecological barrier checking the eastward expansion of the Thar desert and protecting the NW plains from dust and aridity.
- **Aravalli Green Wall Project:** plant native species, restore surface water bodies and rejuvenate degraded land along the range - a domestic mirror of the African GGW's nature-based approach.

• **India's land-degradation commitments:** restore 26 million hectares of degraded land by 2030 (announced at UNCCD COP14, New Delhi, 2019); ties to the Bonn Challenge.

■ MUST-KNOW FACTS

1. Aravalli Green Wall Project launched 2023 (MoEFCC); ~1,400 km along the Aravallis + ~5 km buffer.
2. Covers Haryana, Rajasthan, Gujarat and Delhi; aim: halt eastward spread of the Thar.
3. Aravallis = among the oldest fold mountains; natural barrier against desertification.
4. India pledged to restore 26 million ha of degraded land by 2030 (UNCCD COP14, 2019).
5. Complements ISRO's Desertification & Land Degradation Atlas findings (~30% land degraded).

■ UPSC TRAPS

- ✗ The Aravalli Green Wall is a copy of China's afforestation, unrelated to Africa's GGW.
- ✓ It is explicitly modelled on the concept of Africa's Great Green Wall - a green buffer to halt desert expansion.
- ✗ The Aravallis are young mountains like the Himalayas.
- ✓ The Aravallis are among the OLDEST fold mountains on earth (relict range), unlike the young Himalayas.

■ MAINS ANGLE

Discuss the Aravalli Green Wall Project as India's answer to desertification, and compare its design and challenges with Africa's Great Green Wall.

■ STUDY LINK

Geography -> Aravallis / Indian grasslands + GS-III desertification (UNCCD, Bonn Challenge); contrast with African GGW (Must-Do tier)

End of Register Notes — UPSC Agent / Copilot CLI